SANDRINE BOISSEL, PhD

OPERATIONS, ANALYTICS & SYSTEMS LEADER | SCALING BIOTECH PROCESSES, TEAMS & DIGITAL SOLUTIONS

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SUMMARY OF QUALIFICATIONS

- **Operations and systems leader** with experience designing and scaling **data-driven workflows, planning systems, analytics platforms, and AI evaluation** in high-growth biotech environments.
- Specializes in translating fragmented operational processes into structured, data-driven systems that **improve decision-making, resource allocation, and cross-functional execution**.
- Proven track record **building executive dashboards and operational analytics systems**, leveraging enterprise data to enable scalable reporting and planning frameworks.
- Experienced in **leading cross-functional transformation initiatives** across technical, finance, and business stakeholders, with a strong focus on operational clarity, automation, and measurable impact.

AREAS OF EXPERTISE

Core Competencies:	Process Design & Standardization | Workflow Automation | Business Analytics & Insights | Program & Portfolio Enablement | Continuous Improvement | Product Operations
Technical Skills:	Python | SQL | Tableau | Power BI | Snowflake | OpenAI API | REST API Integration | Git | Jira | Confluence | Smartsheet
Domain Expertise:	Life Sciences | Biotech Operations | Digital Transformation | AI Evaluation | Computational Biology | Molecular Biology | Business Operations
Languages:	English (Fluent) | French (B2) | Spanish (B1)

PROFESSIONAL EXPERIENCE

INDEPENDENT CONSULTANT | Remote | Scientific & Data Consultant **Aug 2025 – Present**

- Contributed to BioSecBench by authoring biology-focused evaluation scenarios for assessing model behavior in dual-use biological contexts.
- Developed expert-level biology prompts and evaluated large language model (LLM) responses for scientific accuracy, factual correctness, and biological reasoning across multiple AI evaluation projects.

GINKGO BIOWORKS, INC. | Boston, MA **Aug 2017 – Aug 2025**

Held progressively senior roles across computational biology, operations, and business systems during Ginkgo Bioworks' growth from startup to public company. Scope expanded from computational design and software-enabled biology into operational leadership and business systems design, ultimately serving as team lead for Business Operations within the largest business unit. Focused on enabling scalable execution through data systems, process design, and cross-functional operational frameworks spanning technical, financial, and business teams.

Associate Director of Operational Excellence | Mar 2023 – Aug 2025

Led operational excellence and business operations for a major business unit, building and scaling data-driven operating systems that improved execution visibility, planning accuracy, and cross-functional alignment across technical, finance, and operations teams.

- Partnered with Finance, Data, and technical leadership to design and implement enterprise planning frameworks supporting 100+ active programs and 200+ contributors, improving forecasting, resource allocation, and cost transparency.

- Developed executive-facing dashboards (Tableau, SQL, Snowflake)—both directly and in partnership with team members—creating a centralized source of operational and performance data used for business decision-making across the organization.
- Redesigned S&OP/S&OE reporting and business review cadences, strengthening the connection between strategic planning and execution through standardized metrics, reporting structures, and recurring decision forums.
- Supported strategic procurement initiatives by building spend and resource visibility dashboards, improving budget transparency and enabling more efficient allocation of shared operational resources.

Operations Lead/Senior Designer I | Jul 2020 – Mar 2023

- Designed and implemented an integrated work planning and execution system for a 22-person team, connecting Jira, Confluence, Smartsheet, Slack, and internal tools to create a unified workflow for prioritization, delivery tracking, and operational visibility.
- Led initiatives to automate data pipelines and modernize computational workflows, improving reliability, scalability, and maintainability of core analytical systems used across the organization.
- Defined and structured new service offerings for the computational biology organization, partnering with Finance, Product, S&OP, and Digital Technology teams to align operational capabilities with evolving commercial and technical requirements.
- Standardized processes and interfaces between internal delivery and customer-facing teams, reducing operational friction and improving cross-functional execution consistency.
- Contributed to the FELIX program (IARPA-funded), supporting development and validation of a sequence analysis pipeline for detecting engineered nucleotide sequences.
- Recognized with a Redwood Award (top ~1.5% performance), awarded for high-impact contributions to operational systems and delivery improvements.

Design Engineer III | Mar 2018 – Jul 2020

Performed computational protein and genetic design alongside software development, contributing to customer projects and internal tooling for high-throughput biological design.

- Contributed to 18 customer projects through computational protein mining, sequence analysis, and candidate identification, including work supporting the development of a commercial food-technology product.
- Developed and enhanced internal computational design software and automation tools, including DNA barcode design systems for error-tolerant encoding/decoding.

Consultant, Lab Operations | Aug 2017 – Feb 2018

- Supported lab build-out operations, including equipment logistics, inventory management, and asset tracking systems for a new 18,000 sq. ft. facility.
- Led procurement and relocation efforts for lab equipment, materials, and personnel, and supported implementation and training for a new inventory management system.

Ginkgo Bioworks, Inc. | Boston, MA | Operations Manager **Jul 2015 – Oct 2015**

Managed various administrative and operational functions, including purchasing, financial accounting and reporting, facilities maintenance, human resources, and space planning.

Harvard Medical School | Boston, MA | Postdoctoral Fellow, Lab of Dr. George Church **Jan 2014 – Jul 2015**

Collaborated with a group of peers (who went on to form GRO Biosciences), performing bacterial genome engineering and directed evolution of aminoacyl tRNA synthetases for incorporation of click-chemistry compatible non-standard amino acids.

Additional Experience

Earlier roles in computational biology, metabolic engineering, and academic research at Harvard Medical School, Old Dominion University, and Greenlight Biosciences focused on genome engineering, bioinformatics, and molecular biology applications.

Education & Certifications

Doctor of Philosophy (PhD), Cellular & Molecular Biology | University of Washington — Received: 2013
Thesis Advisors: Dr. Andrew M. Scharenberg & Dr. David Baker

Master of Science, Marine Biology | Northeastern University — Received: 2017

Certificate in Biomedical & Clinical Studies, Biomedical Sciences | Boston University — Received: 2007

Bachelor of Science, Biology | McGill University — Received: 2005